

# Improve ex-post anyprice-share approximation in best-of-both-worlds chore allocation mechanisms

ChatGPT (gpt-5.6-sol)\*

Partial progress generated on 2026-08-24

## Abstract

*Status: partial progress.* The boundary defects are repaired, and the lower bound is strengthened to hold with identical costs, but the fundamental interval  $[3/2, 2)$  remains unresolved. This note therefore does not constitute a complete solution.

## 1 The open problem

This paper addresses an open problem posed by Uriel Feige, Xin Huang in *On picking sequences for chores* [1] (Economics and Computation (EC), 2023), Section 1.6 "Discussion and problems", page 14 (arXiv:2211.13951v1 pagination). The website catalogue entry is problem 2 (W4383533279\_p0) of the [Open Problems in Operations Research](#) collection.

*Determine whether there exists an allocation mechanism for indivisible chores with additive disvaluation functions and arbitrary responsibilities  $(b_i)_{i=1}^n$  that simultaneously satisfies:*

$$\mathbb{E}[c_i(A_i)] \leq b_i c_i(M) \quad \text{for every agent } i,$$

*and an ex-post APS guarantee with approximation factor strictly smaller than 2, i.e.,*

$$c_i(A_i) \leq \rho \text{APS}(c_i, b_i) \quad \text{for every agent } i \text{ with some constant } \rho < 2.$$

*(Here  $A = (A_1, \dots, A_n)$  is the allocation produced by the mechanism, and the expectation is over the mechanisms internal randomness.)*

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

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\*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve  $\rightarrow$  A.8 verify  $\rightarrow$  A.10 repair write-up; run `run_20260824_024352_9514b3fc, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

## 2 Setting and main result

For each agent  $i$ , let

$$a_i = \text{APS}(c_i, b_i), \quad B_i = b_i c_i(M),$$

and let  $\rho^*$  denote the optimal universal simultaneous ex-ante proportional and ex-post APS factor. The strongest result proved here is the following.

**Theorem 1.**

$$\frac{3}{2} \leq \rho^* \leq 2.$$

The lower bound is established in Section 4 and the upper bound in Section 5; Section 6 records an exact coupling condition and isolates the obstruction that leaves the interval  $[3/2, 2)$  open.

## 3 Conventions and the APS lottery lemma

For  $M = \emptyset$ , every APS is defined to be 0; the unique empty allocation solves the instance. For nonempty  $M$ ,

$$\text{APS}(c, b) = \max_{p \in \Delta(M)} \min\{c(S) : p(S) \geq b\}.$$

For  $T \geq 0$ , put

$$\mathcal{F}_T = \{S \subseteq M : c(S) \leq T\}.$$

**Lemma 2** (APS lottery lemma). *For  $T \geq 0$ ,  $\text{APS}(c, b) \leq T$  if and only if there is a lottery over  $\mathcal{F}_T$  in which every chore has marginal probability at least  $b$ .*

*Proof.* By definition,

$$\text{APS}(c, b) \leq T \iff \min_{p \in \Delta(M)} \max_{S \in \mathcal{F}_T} p(S) \geq b,$$

and the finite minimax theorem gives

$$\min_{p \in \Delta(M)} \max_{S \in \mathcal{F}_T} p(S) = \max_{\mu \in \Delta(\mathcal{F}_T)} \min_{e \in M} \Pr_{\mu}[e \in S].$$

Combining the two displays proves the claim.  $\square$

Applying the lemma at  $T = \text{APS}(c, b)$ , and noting that  $\mathcal{F}_T$  is downward closed—so excess marginals can be thinned independently, producing exact marginals  $b$ —yields

$$b\mathbf{1}_M \in \text{conv}\{\mathbf{1}_S : c(S) \leq \text{APS}(c, b)\}.$$

For  $b > 0$ ,

$$a \geq d \doteq \max_e c(e), \quad a \geq B \doteq b c(M).$$

The first inequality follows by concentrating the prices on a chore of largest cost; the second, when  $c(M) > 0$ , by taking prices proportional to costs (the zero-total-cost case is trivial). If  $a = 0$  and  $b > 0$ , the first inequality implies  $c \equiv 0$ ; thus a zero APS never requires division by the APS.

## 4 A $3/2$ lower bound with identical costs

**Proposition 3.**  $\rho^* \geq \frac{3}{2}$ .

*Proof.* Fix any  $\rho < 3/2$  and choose

$$0 < \varepsilon < \min\{1, 3 - 2\rho\}, \quad L = 2 - \varepsilon.$$

Take three agents with

$$(b_1, b_2, b_3) = \left(\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\right)$$

and four chores whose costs, identical for every agent, are

$$(L, 1, 1, 1).$$

For agents 1 and 2, we claim  $a_1 = a_2 = 2$ . For the upper bound, select  $\{L\}$  with probability  $2/5$  and each of the three pairs of unit chores with probability  $1/5$ : every chore then has marginal  $2/5$ , and every supported bundle costs at most 2. Conversely, under uniform prices  $1/4$ , any set of price at least  $2/5$  contains at least two chores, whose minimum possible cost is 2.

For agent 3, we claim  $a_3 = L$ . Select each singleton with probability  $1/5$  and the empty set with probability  $1/5$ . For the reverse inequality, price the  $L$ -chore at  $1/2$  and each unit chore at  $1/6$ ; any set of price at least  $1/5$  either contains the  $L$ -chore or contains at least two unit chores, so its cost is at least  $L$ .

Moreover,

$$B_3 = \frac{L+3}{5} = 1 - \frac{\varepsilon}{5} < 1,$$

while

$$2\rho < L + 1.$$

Suppose some  $\rho$ -capped allocation gave agent 3 cost below 1. Then agent 3's bundle would be empty, and agents 1 and 2 would have to split all four chores. The agent receiving the  $L$ -chore could not also receive a unit chore, since that would cost  $L+1 > 2\rho$ ; the other agent would then receive all three unit chores, at cost

$$3 > L + 1 > 2\rho,$$

which is impossible. Hence every  $\rho$ -capped allocation satisfies

$$c_3(A_3) \geq 1 > B_3,$$

and no lottery over such allocations can be ex-ante proportional. Therefore  $\rho^* \geq 3/2$ .  $\square$

*Remark.* The endpoint is genuinely strict: at  $\rho = 3/2$ , let agent 3 receive each singleton with probability  $1/5$  and be empty with probability  $1/5$ , and conditionally split the remaining chores symmetrically between agents 1 and 2. Their loads never exceed 3, and every chore has ownership marginals  $(2/5, 2/5, 1/5)$ .

## 5 A universal factor of 2

**Theorem 4.**  $\rho^* \leq 2$ .

*Proof.* Start from the proportional fractional assignment

$$x_{ie} = b_i.$$

Agents with  $b_i = 0$  receive no slots and hence, ultimately, no chores.

For each  $i$  with  $b_i > 0$ , sort the chores in nonincreasing order of  $c_i$ -cost and place their fractional pieces, each of mass  $b_i$ , consecutively into unit-mass slots  $k = 1, \dots, K_i$ . Let  $F_{ik}$  denote the fractional cost in slot  $k$ , and let  $q_{ik}$  be the largest cost of a chore incident to that slot.

The resulting chore–slot edge weights form a fractional bipartite matching: every chore has degree

$$\sum_i b_i = 1,$$

and every slot has degree at most one. Decompose this matching into integral matchings. Every matching of positive probability saturates every chore, because each chore’s integral degree is at most one while its expectation equals one. Projecting from slots to agents therefore yields a lottery over genuine partitions with

$$\Pr[e \in A_i] = b_i;$$

consequently,

$$\mathbb{E}[c_i(A_i)] = B_i.$$

In every realized allocation, at most one chore is selected from each slot. Writing  $d_i = \max_e c_i(e)$ , we have  $q_{i1} = d_i$ ; for  $k \geq 2$ , the preceding slot is full and contains only chores at least as costly as those entering slot  $k$ , so  $q_{ik} \leq F_{i,k-1}$ . Thus

$$c_i(A_i) \leq \sum_k q_{ik} \leq d_i + \sum_{k < K_i} F_{ik} \leq d_i + B_i \leq 2a_i. \quad \square$$

The same construction yields the following rigorous conditional improvement.

**Corollary 5.** *If  $\min\{d_i, B_i\} \leq a_i/2$  for every  $i$ , then every realized allocation above satisfies  $c_i(A_i) \leq \frac{3}{2}a_i$  for every  $i$ .*

*Proof.* The other member of  $\{d_i, B_i\}$  is at most  $a_i$ .  $\square$

*Remark.* When  $a_i > 0$ , one may state the instance-dependent ratio  $(d_i + B_i)/a_i$ . When  $a_i = 0$ , the separate argument above gives realized cost 0; no undefined quotient is used.

## 6 An exact coupling condition and the unresolved gap

Let

$$\mathcal{P}_\rho = \{A = (A_1, \dots, A_n) : A \text{ partitions } M, c_i(A_i) \leq \rho a_i \ \forall i\},$$

and adopt the convention

$$\min_{A \in \emptyset}(\cdot) = +\infty.$$

**Proposition 6.** *An ex-ante proportional lottery supported on  $\mathcal{P}_\rho$  exists exactly when*

$$\min_{A \in \mathcal{P}_\rho} \sum_i \lambda_i c_i(A_i) \leq \sum_i \lambda_i B_i \quad \text{for all } \lambda \in \mathbb{R}_{\geq 0}^n.$$

*Proof.* For nonempty  $\mathcal{P}_\rho$ , this follows from

$$\min_{\mu \in \Delta(\mathcal{P}_\rho)} \max_i (\mathbb{E}_\mu[c_i(A_i)] - B_i) = \max_{\lambda \in \Delta(n)} \min_{A \in \mathcal{P}_\rho} \sum_i \lambda_i (c_i(A_i) - B_i).$$

If  $\mathcal{P}_\rho = \emptyset$ , the  $+\infty$  convention correctly declares the condition false. □

A tempting 3/2 repair remains incomplete. After normalizing  $a_i = 1$ , let

$$H_i = \{e : c_i(e) > 1/2\}.$$

Every unit-cost APS configuration contains at most one member of  $H_i$ , so

$$b_i |H_i| \leq 1.$$

Thus the large-edge vector  $z_{ie} = b_i$ , supported on the pairs  $(i, e)$  with  $e \in H_i$ , is a fractional bipartite matching and can be rounded integrally. However, conditioning agent  $i$ 's APS lottery on the chosen large chore changes its residual small-chore marginals; once different agents have been conditioned, the residual marginals need not sum to one for each chore. The residuals therefore need not form a fractional partition, and small-chore slot rounding cannot yet be applied. This is the precise unresolved simultaneous-coupling gap.

Accordingly, the fundamental question remains open:

$$\frac{3}{2} \leq \rho^* \leq 2,$$

and neither a universal constant below 2 nor an impossibility for every  $\rho < 2$  has been proved here.

## References

- [1] Uriel Feige, Xin Huang. *On picking sequences for chores*. Economics and Computation (EC), 2023. <https://doi.org/10.1145/3580507.3597783>